Conditional Laguerre Clustering and Mixed Cardinal Tests for the Theta Form

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November 2026
Research draft; independent review pending

Abstract

For a hypothetical nonreal zero of the Riemann xi function we compare, pair by pair, its differentiated Hadamard kernel with a positive Poisson kernel. On the complement of the unit disk this sharpens a previous lower coefficient from 1/3 to 1; under an additional Laguerre-sign assumption it also places a neighboring zero in a positive-contribution cone. Assuming the stated cardinal-source realization, we give exact mixed tests for two nonreal quartets and for one real and one nonreal channel. A further conditional real-neighbor consequence requires an independent global polynomial-form positivity hypothesis and an explicit quantitative inequality. Neither that positivity nor the Laguerre sign is proved here. No off-critical zero is asserted to exist and no conclusion resolves the Riemann hypothesis. The status of historical priority for the estimates is not determined.

1 Xi zeros, the positive budget, and the sign assumption

Put $F(z) = \Xi(z) = \xi(1/2 + iz)$ in the standard normalization [4]. Its distinct zero coordinates are denoted $\lambda = a + ib$, each with multiplicity n_λ, and lie in $|b| < 1/2$; they are symmetric under conjugation and negation. In this paper a *target* is a hypothetical zero $w = t + iy$ with $t > 0$, $0 < y < 1/2$ and multiplicity ν. Let m be the full multiplicity of a real zero at t, or zero if none, and write $G_t(z) = F(z)/(z-t)^m$. Thus $G_t(t) \neq 0$ and

$$D(t) = \frac{G_t'(t)^2 - G_t(t)G_t''(t)}{G_t(t)^2} = -\frac{2\nu}{y^2} + \sum_{\lambda \in Z_*} n_\lambda \operatorname{Re} \frac{1}{(t-\lambda)^2}. \tag{1}$$

Here Z_* removes the center t and $t \pm iy$, each with full multiplicity; in particular the reflected pair $-t \pm iy$ remains. The sum is absolutely convergent by the zero count and the differentiated Hadamard product. This is the generalized Laguerre coefficient ratio at a center zero, not necessarily the ordinary first Laguerre sign. We will explicitly assume $D(t) \geq 0$ only where stated; it is not established for hypothetical targets [1].

Set

$$A(t) = \operatorname{Re} \frac{\xi'}{\xi}(3/2 + it), \quad P_t(a+ib) = \frac{1+b}{(t-a)^2 + (1+b)^2}, \quad A_* = A(t) - m - \frac{2\nu}{1-y^2}. \tag{2}$$

The absolutely convergent paired Hadamard sum gives $A(t) = \sum_\lambda n_\lambda P_t(\lambda)$, so $A_* = \sum_{Z_*} n_\lambda P_t(\lambda) > 0$. The corresponding absolutely convergent prime-power representation and $A(t) = \frac{1}{2}\log(t/(2\pi)) + O(1)$ as $t \to +\infty$ are given in [1]. No numerical value of A_* at an unknown target is asserted.

2 Pair comparison and positive-cone clustering

For a pair $a \pm ib$ in Z_* put $\delta = t - a$, $h = \delta^2 + b^2 > 0$ and $v = b^2 \le 1/4$. The ratio of the signed pair contribution in (1) to the pair contribution to A_* is

$$R(h, v) = \frac{(h - 2v)((h + 1)^2 - 4v)}{h^2(h + 1 - 2v)}. \tag{3}$$

For a real zero take instead $R(h, 0) = 1 + 1/h$. The common multiplicity cancels in the ratio.

Lemma 2.1. *For $h > 0$ and $0 \le v \le 1/4$, $R(h, v) \le 1 + 1/h$. If $h \ge 1$, then $1 \le R(h, v) \le 2$. The lower constant 1 is sharp as a universal pair-kernel constant.*

Proof. The denominator in (3) is positive and direct simplification gives

$$1 + \frac{1}{h} - R = \frac{2v(3h + 1 - 4v)}{h^2(h + 1 - 2v)} \ge 0.$$

For $h \ge 1$ the numerator of $R-1$ is $h^2+h-8hv+8v^2-2v$. With $u = h-1 \ge 0$ and $z = 1-4v \ge 0$ this is $u^2 + 2uz + u + z^2/2 + 3z/2 \ge 0$. The upper bound follows from $1 + 1/h \le 2$; $R(h, v) \to 1$ as $h \to \infty$. □

Let S_{loc} be the signed sum over $\lambda \in Z_*$ with $|t - \lambda| < 1$, and let A_{far} be the P_t sum over its complement, both with multiplicities. This local set is finite and conjugation invariant. Lemma 2.1 yields

$$-\frac{2\nu}{y^2} + S_{\mathrm{loc}} + A_{\mathrm{far}} \le D(t) \le -\frac{2\nu}{y^2} + S_{\mathrm{loc}} + 2A_{\mathrm{far}}. \tag{4}$$

The lower coefficient improves the 1/3 coefficient of the pointwise far bound in [1] by pairing conjugates before comparison. Formula (4) does not supply the local zero data or certify $D(t)$.

Theorem 2.2 (A neighboring zero in the positive cone)**.** *Assume $D(t) \ge 0$. Let $Z_+ = \{a + ib \in Z_* : (t - a)^2 > b^2\}$ and $d_+ = \min_{\lambda \in Z_+} |t - \lambda|$. Then Z_+ is nonempty and the minimum exists. If $A_* y^2 < 2\nu$, then*

$$d_+^2 \le \frac{A_* y^2}{2\nu - A_* y^2} =: r^2. \tag{5}$$

In particular the neighbor has real part different from t. When $A_* y^2 < \nu$, the right side of (5) is less than 1 and is no larger than $2A_* y^2/(3\nu - A_* y^2)$, the preceding distance bound in [1].

Proof. The signed kernel of an individual zero has the sign of $(t-a)^2-b^2$; deleting all nonpositive terms from (1) gives positive sum at least $2\nu/y^2$. Hence Z_+ is nonempty. Discreteness of the zeros makes its distance minimum attained. Pairing the positive terms and applying Lemma 2.1, their sum is at most $(1 + d_+^{-2})A_*$. Solving $2\nu/y^2 \le (1 + d_+^{-2})A_*$ proves (5). The final comparison is immediate when $A_* y^2 < \nu$. □

The neighbor may be real or nonreal; in the general bound it may be the reflected target pair. By Platt and Trudgian's verified height [5], a hypothetical target has $t > 3 \cdot 10^{12}$. Thus when $r < 1$ the reflected pair is not that neighbor. The verification does not imply that the neighbor is real or lies below the verified height.

If separately $d_+ \ge d_0 > 0$, (5)'s preceding inequality implies $y^2 \ge 2\nu d_0^2/[A_*(1+d_0^2)]$. Under the independent input $I_w \ge y^2/200$ stated in Section 4, it follows that $4\nu I_w \ge \nu^2 d_0^2/[25A(t)(1 + d_0^2)]$. Its right side need not have a uniform positive lower bound as $t \to \infty$.

3 The centered realization and cardinal inputs

Write $r = |x|$ and define

$$\kappa(x) = \sum_{n\geq 1} (4\pi^2 n^4 e^{9r/2} - 6\pi n^2 e^{5r/2}) e^{-\pi n^2 e^{2r}}, \qquad W(x) = \frac{\kappa(x)}{2\cosh(x/2)}.$$

The weight is positive and even, and $\int \kappa(x) e^{izx} dx = \Xi(z)$ in this normalization. Let $\mathcal{H} = L^2_{\mathrm{even}}(\mathbb{R}, dx/W)$, $m(u) = \int \cosh(x/2) u(x) dx$, and $\mathcal{H}_0 = \ker m$. Inner products are linear in the first variable; unmarked integrals are over $\mathbb{R}$ and $q(u, v)$ denotes the polarized form. On the centered minimal form domain $\mathcal{X}_0$, the closure of centered even compact smooth functions in the squared norm $\|u\|^2_{\mathcal{H}} + t[u]$, set

$$t[u] = \int_0^\infty \frac{e^{-s/2}}{1 - e^{-2s}} \|u(\cdot + s) - u\|^2_{L^2(dx)}\, ds,$$
$$q[u] = t[u] - (\log \pi - \psi(1/4))\|u\|^2_{L^2(dx)} - 2\sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} \operatorname{Re} \int u(x + \log n)\overline{u(x)}\, dx.$$

We take as *inputs*, not reprove here, the closed form realization and the following actual cardinal-source identities of [2]. Its associated self-adjoint L acts in $\mathcal{H}_0$, with all prime powers and the full Gamma contribution retained. Put $K = \ker L$ and $Q = I - P_K$. For $u \in \operatorname{Dom} L$, $Qu \in \operatorname{Dom} L$, $LQu = Lu$, and $q[Qu] = q[u]$. No identification of K with a known derivative span is assumed.

For an actual zero w of multiplicity ν, one has $w \neq 0$ and $w^2 \neq -1/4$; define the removable entire quotient

$$a_w = \frac{\Xi^{(\nu)}(w)}{\nu!(2w)^\nu}, \quad H_w(z) = \frac{\Xi(z)}{a_w}\left[\frac{1}{(z^2 - w^2)^\nu} - \frac{1}{(-1/4 - w^2)^\nu}\right], \quad \widehat{h}_w = H_w.$$

The stated realization input includes $h_w \in \operatorname{Dom} L$, the cardinal values and, with $g_w = W\cos(\bar{w}x)$, $Lh_w = 2\nu g_{\bar{w}}$. For distinct nonreal quartets, $h_{i,-} = \operatorname{Im} h_{w_i}$ and $e_i = Qh_{i,-}/\sqrt{\nu_i}$ obey

$$q(e_i, e_j) = -\delta_{ij}, \qquad Le_i = -2\sqrt{\nu_i}\, W D_i, \quad D_i(x) = \sin(t_i x)\sinh(y_i x). \tag{6}$$

For a distinct real zero $a > 0$ of multiplicity m_a, the real cardinal h_a satisfies $q[h_a] = 2m_a$, $Lh_a = 2m_a W\cos(ax)$, and is q-orthogonal to the negative channels. The same equalities hold with h_a replaced by Qh_a. These assumptions are stronger than mere formal Fourier division; domain admission and the exact image are essential.

For $\epsilon > 0$ define on $\operatorname{Dom} L$ the real quadratic form

$$\Phi_\epsilon[u] = \|Lu\|^2_{\mathcal{H}} + \epsilon q[u]. \tag{7}$$

Where used below, global $\Phi_\epsilon[u] \geq 0$ for all $u \in \operatorname{Dom} L \cap K^\perp$ is a separate hypothesis. It is not proved and must not be inferred from any consequence derived using it. By the spectral theorem, it is equivalent to $\sigma(L) \cap (-\epsilon, 0) = \varnothing$; it does not by itself imply $q \geq 0$ and allows negative spectrum at or below $-\epsilon$.

4 Two exact mixed tests

Let $w_i = t_i + iy_i$ be two distinct hypothetical nonreal quartets with positive t_i, y_i and multiplicities ν_i. From (6), with $c = (\sqrt{\nu_2}, -\sqrt{\nu_1})/\sqrt{\nu_1 + \nu_2}$,

$$\Phi_\epsilon[c_1 e_1 + c_2 e_2] = \frac{4\nu_1\nu_2}{\nu_1 + \nu_2} \int_{\mathbb{R}} W|D_1 - D_2|^2 dx - \epsilon. \tag{8}$$

The vector has Euclidean coefficient norm one in the displayed q-normalized coordinates, not necessarily unit $\mathcal{H}$ norm. A negative right side is an admitted negative test direction for Φ_ϵ, so it disproves global nonnegativity. The corresponding ratio of coefficients of the unnormalized $Qh_{i,-}$ is $\nu_2 : -\nu_1$. Passing the two individual scalar tests does not establish (8) is nonnegative.

Define finite constants $M_j = \int W x^j \cosh^2(x/2)dx$ for $j = 2, 4$ and $Y = \max(y_1, y_2)$. Mean-value estimates, $|\sin v| \le |v|$ and Minkowski give

$$\left(\int W|D_1 - D_2|^2 dx\right)^{1/2} \le Y|t_1 - t_2|\sqrt{M_4} + |y_1 - y_2|\sqrt{M_2}. \tag{9}$$

Indeed, $|\sin(t_1x) - \sin(t_2x)| \le |t_1 - t_2||x|$, $|\sinh(y_1x)| \le Y|x|\cosh(x/2)$, and $|\sinh(y_1x) - \sinh(y_2x)| \le |y_1 - y_2||x|\cosh(x/2)$. Hence the right side of (8) is negative if $4\nu_1\nu_2/(\nu_1+\nu_2)$ times the square of the right side of (9) is strictly smaller than ϵ. This is conditional on two actual quartets and their parameters; Theorem 2.2 does not establish a nonreal neighbor or closeness of imaginary parts.

For a distinct real zero $a > 0$ of multiplicity m_a, put $E_a = \cos(ax)$, $D_w = \sin(tx)\sinh(yx)$ and

$$K_a = \int W E_a^2 dx, \quad J_a = \int W E_a D_w\, dx, \quad I_w = \int W D_w^2 dx.$$

In the coordinates $Qh_a/\sqrt{2m_a}$ and $Qh_{w,-}/\sqrt{\nu}$, the exact Hermitian matrix of (7) is

$$\begin{pmatrix} 2m_aK_a + \epsilon & -2\sqrt{2m_a\nu}\, J_a \\ -2\sqrt{2m_a\nu}\, J_a & 4\nu I_w - \epsilon \end{pmatrix}. \tag{10}$$

The first diagonal entry is strictly positive. Consequently, this two-dimensional restriction has a negative direction if and only if

$$(2m_aK_a + \epsilon)(4\nu I_w - \epsilon) < 8m_a\nu J_a^2. \tag{11}$$

This determinant test needs both an actual real zero and a separate verification of (11); it does not prove either. We also record $I_w \le y^2 M_2$. The additional lower bound $I_w \ge y^2/200$ for $t \ge 10^6$ and $0 < y \le 1/2$ is a separate input from the rational-enclosure argument in [3]; we do not re-certify that calculation here.

5 A conditional real-neighbor consequence and its limit

Corollary 5.1. *Let a target as in Section 1 satisfy $D(t) \ge 0$ and $A_*y^2 < \nu$, and let r be defined by (5). Assume also $\Phi_\epsilon \ge 0$ on $\mathrm{Dom}\, L \cap K^\perp$ and $\epsilon \ge 2A_*M_2r^2$. Then there is a real zero $a > 0$ in the positive cone with $0 < |t - a| \le r$.*

Proof. By Theorem 2.2, a neighbor satisfies $h \le r^2 < 1$ and $\delta^2 > b^2$. If $b \ne 0$, then $b^2 < h/2 \le r^2/2$. The pair contribution to A_* per unit multiplicity is

$$\frac{2(1 + h - 2b^2)}{(1+h)^2 - 4b^2} \ge 1,$$

because the difference of numerator and denominator is $1 - h^2 \ge 0$. Thus the neighbor's multiplicity ν_2 is at most A_*. Since $t > 3 \cdot 10^{12}$ by [5], its real coordinate is positive and it is not the reflected target; it defines a distinct nonreal quartet. The scalar test from (6) and global $\Phi_\epsilon \ge 0$ require $\epsilon \le 4\nu_2 I_2 \le 4A_*M_2b^2 < 2A_*M_2r^2$, a contradiction. Hence $b = 0$ and the cone condition gives $a \ne t$. □

The hypothesis is a necessary-condition framework, not a proof of global positivity. The target scalar direction already forces $\epsilon \le 4\nu I_w$; therefore the inequalities assumed in Corollary 5.1 can coexist only if $2A_* M_2 r^2 \le 4\nu I_w$. The independent lower bound $I_w \ge y^2/200$ makes $2A_* M_2 r^2 \le \nu y^2/50$ a sufficient check for this *one* compatibility inequality, not for global positivity or actual existence of the assumed zero configuration. Conversely, any proposed sufficient separation condition that already forces $\epsilon > 4\nu y^2 M_2 \ge 4\nu I_w$ contradicts the target scalar condition and makes the global-positivity implication vacuous. The conclusion must not be used to prove the hypothesis from which it was derived.

6 A model and the scope of the result

For $t > 1$ and $0 < y < 1/2$, the even real polynomial

$$H_y(z) = \big((z-t)^4 - y^4\big)\big((z+t)^4 - y^4\big)$$

has target zeros $t \pm iy$ and real neighbors $t \pm y$. At t the first four roots contribute $-2/y^2 + 2/y^2 = 0$ to the differentiated logarithmic sum; the four reflected roots contribute positively because their squared horizontal separation exceeds y^2. Thus its analog of $D(t)$ is positive, while $y \to 0$ and the remaining positive Poisson budget stays bounded. This is a comparison model, not an example of xi zeros or of the prime explicit formula. It is a special case of an earlier boundary-cluster concealment model already considered in related working notes; it is a rediscovery, not a claim of novelty for that model. We make no claim of historical priority for the present inequalities or for classical Hadamard, Laguerre, or Schur identities.

The conclusions involving $D(t)$ and Φ_ϵ are conditional on different unproved signs. The exact source tests additionally depend on the stated operator and domain inputs. No off-critical zero has been exhibited here, no actual zero configuration is numerically certified, and neither the Riemann hypothesis nor its negation is established.

References

[1] J. Choi, *Actual-theta Laguerre signs: local zero clusters and a prime-controlled tail*, research draft, 4 November 2026, PDF SHA-256 `e637b78605a377ede1fb09ec9408fcb656a46f2fb66cfee0c37bd962276cf003`. The supplied working manuscript is an analytic input; its public availability is not assumed.

[2] J. Choi, *Cardinal theta sources and the sign on the full kernel complement*, research draft, 4 November 2026, source archive SHA-256 `c0d584c3cbb416e962479a2f919ec9a92941add97cdf4dff002f303b94b2fbea`. Its realization and cardinal identities are assumed here.

[3] J. Choi, *A certified two-sided theta integral bound and the remaining zero-distance condition*, research draft, 4 November 2026, source archive SHA-256 `29f63ba8bc2fc4ef6f65811586891fe0fd27ce9393503a6778110a96bf154b50`. Its rational enclosure is an independent input here.

[4] NIST Digital Library of Mathematical Functions, §§25.2(iv), 25.4, 25.10(i), https://dlmf.nist.gov/25.2.iv, https://dlmf.nist.gov/25.4, https://dlmf.nist.gov/25.10.i.

[5] D. J. Platt and T. S. Trudgian, The Riemann hypothesis is true up to $3 \cdot 10^{12}$, *Bull. Lond. Math. Soc.* 53 (2021), 792–797, https://doi.org/10.1112/blms.12460; https://arxiv.org/abs/2004.09765.